RK500-23 Soil EC& Salinity Sensor

The probe of Salinity sensor is made of graphite electrode that has the characteristics of stable performance,high sensitivity, wide application scope. Salinity sensor is of simple structure, with stable performance, easy for operation, used in the field monitoring of water & salt dynamics in soil.Therefore,it is an ideal observation instrument in the study of generation,evolution,and improvement utilization of saline soil.Also,it can be used in the anti-corrosion monitoring of underground oil,gas pipelines and other pipelines.The sensor an also be directly submerged in liquid,to measure its electrical conductivity.

FEATURES

- On-line & real-time measurement
- With temperature compensation
- High accuracy
- Simple operation and high reliability
- Fast response
- Strong in corrosion resistance

Parts:

1. Sensor: 1
2. Bracket(optional): 1

SPECIFICATIONS

Item	Technical Specification
Range	0～1000us/cm, 0～2000us/cm,0～5000us/cm,0～10000us/cm 0-20000us/cm
Supply	Mark on the label
Accuracy	±1%
Response time	<1s
Output Signal	4-20mA,RS485
Electrode	316L&Graphite
Principle	Frequency conversion method
Temperature compensation	Automatic(0 -150 °C)
Operating Temperature	0-50℃
Ingress Protection	IP68
Storage	10-60℃@20%-90%RH

MOUNTING

In soil

Section installation

In the area of need to measure dig a soil profile, determine the position of the sensor and depth in the section, the section on level to dig deep for 10-20 cm round hole, insert the sensor level until the bottom of the hole, then filled out of the clay compaction, to ensure that the sensor electrode surface closely contact with soil.Sensors buried, the soil pit backfill compaction sequence according to the original soil layer.

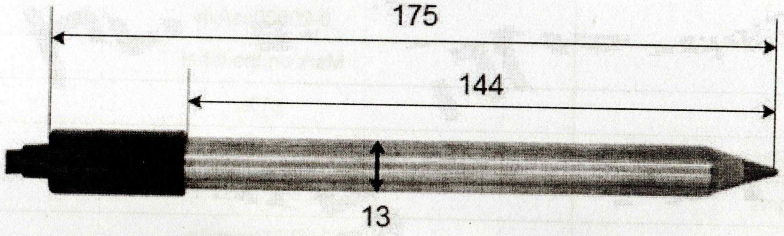
Ground stiletto installation

In the measurement area, down from the ground to play a hole to the desired depth, sensor inserted to the bottom of the hole, closely contact with the soil, will then stratified backfill compaction.

Measure the soil solution

Take a 1 mm sieve separation of 20.00 g dried soil, put in 250 ml bottle drying triangle, add distilled water 100 ml (water:soil = 5:1), oscillation for 5 minutes, the filtered in dry triangle in the bottle.Take the filtered solution 30 ml, placed in small 50 ml beaker,   the sensor electrode in the leaching liquid to be tested.

PRODUCT SIZE



ELECTRICAL CONNECTIONS

Cable(Connector)	Current	Voltage	RS485
Red(Pin1)	V+	V+	V+
Black (Pin4)	V-	V-	V-
White	Signal out	Signal out	
Yellow(Pin2)			RS485A
Green(Pin3)			RS485B

Note: This product has been tested and complies with European CE requirements for EMC directive.

Communication Protocol（MODBUS）

Transmission mode: MODBUS-RTU,**Baud rate:** 9600bps,**Data bits:**8,**Stop bit:**1,**Check bit:**no

Slave address:the factory default is 04H (set according to the need,01H to F7H)

- **The 03H Function Code Example: Read The EC Value(Data type is floating point)**

Host Scan Order(slave address:0x04)

04 <u>03 00 00 00 0A</u> C598

Slave Response

04 <u>03 14</u> 3F 82 DC 81 41 1C 80 64 41 DC F9 0C 43 FF 96 AC 44 0C 92 DF E1 37

EC:（3F82DC81）>>1.022 mS/cm

Salinity:(440C92DF) >>562PPM

- **The 06H Function Code Example: Modify the slave address**

Host Scan Order (Changed the 0CH to 01H):

0C <u>06 00 14 00</u> 01 0913

Slave Response:

0C <u>06 00 14 00</u> 01 0913

Note:

1. All underlined is fixed bit;

2. The last two bytes is CRC check command.

d